

Mayank Bhati

Backend + Applied AI Engineer • Distributed Systems • Java / Spring Boot • Forward Deployed Engineering

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PROFESSIONAL SUMMARY

Backend engineer with 4+ years building and operating large-scale distributed systems in 24/7 production at Adobe and Samsung. Owned RosterSync components serving **6.5M+ K-12 users**, cutting worst-case sync time by **~85%** and designing heartbeat-based recovery for long-running jobs. Built the **RosterSync Incident Copilot** as an internal Adobe hackathon prototype, applying RAG and LLMs to combine live operational evidence with retrieved engineering knowledge.

WORK EXPERIENCE

Adobe | Member of Technical Staff 2 (Backend)

Jun 2024 - May 2026

- **Owned core components of a 4-service, event-driven roster reconciliation platform** (Java 17, Spring Boot, Kafka, Redis, PostgreSQL/Aurora) keeping Adobe Admin Console groups consistent with Clever and ClassLink for **6.5M+ K-12 users at 99.9%+ uptime**.
- **Cut worst-case sync for a 120k+ user district from ~12.5h to ~1h50m (~85%)** through bounded-parallel chunk processing, parallel batch writes, Aurora read replicas, and JVM/GC tuning - without increasing infrastructure spend.
- **Designed self-healing execution for long-running jobs** using task-level heartbeats and atomic stale-work claiming after ephemeral pod crashes; fixed Redis/state drift through event-driven cache invalidation.
- **Re-architected roster enrichment from O(N) per-user downstream calls to bulk aggregation** with Redis-backed per-user builders, reducing downstream calls by **~99.99%** and eliminating ClassLink HTTP 429 throttling.
- **Drove production reliability across 4 services** using Splunk/Grafana, alerting, probes, Resilience4j, Spring Retry, idempotent processing, and bounded error payloads; MTTD <15 min; resolved **150+ Critical/Blocker issues**.

SELECTED INTERNAL PROJECT - ROSTERSYNC INCIDENT COPILOT (ADOBE HACKATHON)

- Built a **Spring Boot + RAG incident-investigation prototype** combining live sync state and logs/metrics with runbooks, architecture docs, error documentation, and historical incidents to produce grounded diagnoses with citations.
- Designed separate live-tool and knowledge-retrieval paths using **embeddings, vector search, hybrid semantic + keyword retrieval, metadata/ACL filtering, reranking, and structured LLM output**; added evaluation for recall, groundedness, citation precision, latency, graceful degradation, and tenant-aware filtering.

Samsung | Software Development Engineer

Jul 2022 - Jun 2024

- Built and operated **catalogue and checkout services** (Java, Spring Boot, JPA/Hibernate, PostgreSQL) sustaining millions of daily requests at 99.9%+ uptime, with order volumes scaling **10x** during Galaxy launches and festive sales.
- Engineered surge handling using bounded thread pools and Kafka-based load leveling, converting spiky bursts into steady, survivable throughput under peak load.
- Reduced **REST API latency by 15-25%** through query-plan analysis, composite-index tuning, N+1 elimination, and targeted Redis caching on catalogue hot paths.
- Hardened checkout with idempotency keys, retry-safe API contracts, circuit breakers, and end-to-end SIT/UAT ownership.

TECHNICAL SKILLS

Languages	Java 17/21, SQL
Backend	Spring Boot, Spring MVC, Spring Security, JPA/Hibernate, JUnit 5, Mockito, REST APIs
Distributed Systems	Microservices, Event-Driven Architecture, Kafka, Concurrency, Idempotency, Backpressure, Caching, Retries, Circuit Breakers, DLQ
Databases & Caching	PostgreSQL, Aurora, Redis, MySQL, MongoDB
AI / LLM Systems	RAG, Embeddings, Vector Search, Hybrid Retrieval, Reranking, Tool Retrieval, Structured Output, Context/Citation Design, LLM Evaluation
Cloud & Observability	AWS, ECS, S3, CloudWatch, Docker, Kubernetes, CI/CD, Git, Maven, Gradle, Splunk, Grafana

EDUCATION

National Institute of Technology, Uttarakhand

2018 - 2022

B.E. Electrical & Electronics Engineering | CGPA 7.99/10